

Silverline House — 25-Storey Residential HRB

Pile Design and Testing Strategy

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1 Introduction

This document forms part of the Gateway 2 application for Building Control Approval submitted to the Building Safety Regulator under the Building Safety Act 2022. The information presented herein has been prepared in accordance with the CLC Guidance Suite for new Higher-Risk Buildings and the requirements set out in Guidance Note 04 (Application Information Schedule) and Guidance Note 06 (Document Management and Submission). All design work described in this document has been carried out by suitably qualified engineers and specialists with relevant experience in higher-risk residential buildings. The foundation system consists of 600 mm diameter continuous flight auger (CFA) piles installed to a minimum depth of 18 m below existing ground level, penetrating at least 3 m into the Lambeth Group. Pile design has been carried out in accordance with BS EN 1997-1 using a combination of ground model analysis and empirical correlations from the site investigation data. The working load per pile is 2,800 kN in compression and 800 kN in tension. Settlement analysis indicates maximum pile head settlements of 8 mm under working loads, with differential settlement between adjacent piles not exceeding 5 mm. Pile load testing in accordance with BS EN ISO 22477-1 is required during construction, with at least two maintained load tests to 150% of the working load prior to commencing pile cap construction.

The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

2 Design Parameters

A site investigation was carried out in accordance with BS EN 1997-2 and BS 5930. The investigation comprised twelve rotary core boreholes to a maximum depth of 30 m below existing ground level, together with in-situ standard penetration tests at 1.5 m intervals. Laboratory testing was carried out on selected samples to determine soil classification, shear strength parameters, consolidation characteristics, and chemical contamination. The ground conditions comprise approximately 3 m of made ground overlying Terrace Gravels (3–9 m depth), London Clay (9–22 m), and Lambeth Group deposits below 22 m. Groundwater was encountered at 2.5 m depth in the gravels. The design groundwater level for structural purposes is taken as 0.5 m below existing ground level. The design has been developed in accordance with the Building Regulations 2010 and the relevant Approved Documents. Where European standards (Eurocodes) are adopted, the applicable UK National Annexes have been used. Design standards and codes of practice referenced in this document include, but are not limited to, BS EN 1990 (Basis of structural design), BS EN 1991 (Actions on structures), BS EN 1992 (Concrete structures), BS EN 1993 (Steel structures), and BS EN 1997 (Geotechnical design). Where proprietary systems or products are specified, the manufacturer's technical data and third-party test evidence are referenced in the relevant sections.

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3 Testing Programme

Static maintained load tests will be carried out on two trial piles prior to commencement of production piling. Each test pile will be loaded to 150% of the working load in accordance with BS EN ISO 22477-1. Dynamic load testing using the Pile Driving Analyser (PDA) method will be carried out on a minimum of 5% of production piles as a quality assurance measure. Sonic integrity testing will be carried out on 100% of production piles in accordance with BS EN ISO 22477-10. All reinforced concrete elements are specified in C32/40 concrete (characteristic cylinder / cube strength) with a maximum water-cement ratio of 0.55 and minimum cement content of 300 kg/m³. Post-tensioned slab tendons are specified in low-relaxation 15.7 mm strand to BS 5896, stressed to 75% of characteristic tensile strength. Reinforcement throughout is Grade B500B to BS 4449. Concrete cover to reinforcement is 35 mm to internal elements and 50 mm to external elements exposed to the weather, in accordance with BS EN 1992-1-1 Table 4.4N for exposure class XC3. Minimum cement content and maximum water-cement ratio comply with the durability requirements of BS 8500-1 for the intended service life of 50 years.

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